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Stacking Deep learning and Machine learning models for short-term energy consumption forecasting

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ABSTRACT

Accurate prediction of electricity consumption is essential for providing actionable insights to decision-makers for managing volume and potential trends in future energy consumption for efficient resource management. A single model might not be sufficient to solve the challenges that result from linear and non-linear problems that occur in electricity consumption prediction. Moreover, these models cannot be applied in practice because they are either not interpretable or poorly generalized. In this paper, a stacking ensemble model for short-term electricity consumption is proposed. We experimented with machine learning and deep models like Random Forests, Long Short Term Memory, Deep Neural Networks, and Evolutionary Trees as our base models. Based on the experimental observations, two different ensemble models are proposed, where the predictions of the base models are combined using Gradient Boosting and Extreme Gradient Boosting (XGB). The proposed ensemble models were tested on a standard dataset that contains around 500,000 electricity consumption values, measured at periodic intervals, over the span of 9 years. Experimental validation revealed that the proposed ensemble model built on XGB reduces the training time of the second layer of the ensemble by a factor of close to 10 compared to the state-of-the-art , and also is more accurate. An average reduction of approximately 39% was observed in the Root mean square error.

1. Introduction

Energy forecasting can be defined as predicting future electricity consumption using historical data. Accurate estimation of electricity consumption can have both economic and environmental benefits. If the electricity consumption demand is known in advance, only the necessary amount of resources could be allotted to satisfy that demand [1]. Efficiently using energy can result in energy savings and by producing only the right amount of energy, the amount of carbon dioxide emitted can be reduced [2,3]. Hence, it is vital to have efficient electric energy management systems and prediction strategies.

Traditional time series forecasting approaches include statistical analysis, Auto-regressive Integrated Moving Average (ARIMA), Seasonal Auto-regressive Integrated Moving Average (SARIMA), Exponential smoothing etc. These approaches assume that the relationship between the items in time series data is linear. However, more often that not, time series forecasting problems are non-linear. Machine learning based models can model the non-linear dependencies in time series data. These models can automatically learn the temporal structures like trends and seasonality. Machine learning based models also tend to be better at generalizing on data that it has not previously seen. The data is re-structured to look like a supervised learning problem. Data from a certain time-period is fed to the model as input. The model uses this data to predict the data in the next time-period.

Several researchers have used Machine learning (ML) and deep neural models for the task of energy consumption forecasting. Support Vector Machine (SVM) was observed to outperform Auto-regressive Moving Average Model (ARMA) and Elman Recurrent Neural Networks for the Mackey–Glass dataset [4]. A multivariate model was proposed to predict stock market prices using Backpropagation networks [5]. A semi-chaotic time series dataset was used which contained non-linear dependencies. Long Short Term Memory (LSTM) was observed to outperform several traditional time series analysis methods, for time series prediction on several finance related datasets [6]. An average reduction of around 85% was observed in the predictions of LSTM as compared to ARIMA.

Ensemble learning performs superior to single regression models [7], as a regression model needs to search over the entire hypothesis space to find the best hypotheses that will fit the data. The hypotheses space of each individual model might not contain the true hypotheses function. However, using ensemble learning, a combination of multiple

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hypotheses is presented, increasing the chances of providing a good approximation of the true hypotheses [8]. A regression model performs a local search to minimize the objective function, while individual models can get stuck in a local optimum. Having multiple models reduces the probability of this happening since the local search is initiated from multiple starting points. Particularly, stacking is generally a better ensemble technique than selecting the best performing base model [9]. Stacking can be used to learn more complicated non-linear dependencies between the input and the output. Divina et al. [10] proposed an ensemble scheme consisting of Random forests, evolutionary trees, and artificial neural networks in their first layer and gradient boosting method in the second layer.

Our work is an extension proposing an improvement over Divina et al.'s [10] approach, but we aim to address some specific limitations that have been observed in existing works. We perform short-term electricity consumption forecasting using stacking ensemble learning by considering energy consumption values from the historical data and using them to predict the consumption values for the next 4 h. Historical window of sizes 16 h and 28 h is considered. We use differencing to remove the seasonality and trend in the dataset, which makes the dataset stationary. Such a preprocessing technique to make the dataset stationary has not been considered in existing works. We consider Deep Neural Networks in place of the shallow neural networks, such that at each hidden layer of the neural network, a more abstract representation of the input is learned, for modeling non-linear dependencies between the input and output compared to a shallow neural network. The major contributions of this work are summarized below:

- We propose a novel ensemble model for short-term energy consumption forecasting, while addressing the limitations of the current approaches, to obtain a more accurate and scalable model.
- To obtain more accurate predictions, we preprocess the dataset using differencing, and the order of differencing is determined by hypotheses testing.
- The hyperparameters of the models used are estimated by a Random Search, so that a larger search space can be explored with the same computational cost.
- An LSTM and XGBoost ensemble is proposed, for much faster and scalable predictions, when compared to the state-of-the-art.
- We benchmark the proposed approach against state-of-art works using standard performance measures.

The remainder of this article is organized as follows: Section 2 presents a discussion on various existing ensemble models for timer series forecasting. Section 3 contains the proposed methodology, and the improvements made to address the limitations of the existing solutions are highlighted. Section 4 presents the results of the experiments conducted to benchmark the proposed ensemble models against the state-of-art works, followed by conclusion and the scope for future work.

2. Related work

Using ensemble learning for time series forecasting is not new to the scientific community. Gabralla et al. [11] proposed an ensemble model with feed-forward neural networks, recurrent neural networks, and radial basis function networks as the base layer to forecast West Texas Intermediate oil prices. The predictions of these base models were combined using the boosting technique. It was observed that the ensemble model outperforms individual base models. Pierdzioch et al. [12] proposed a real-time boosting approach for forecasting gold price fluctuations. The errors of the predictions were observed to be very low. However, the authors have not compared the results with any other model.

Yu et al. [13] proposed a decomposed and ensemble learning model and extreme learning machine to forecast crude oil prices. This ensemble model was observed to perform better than ARIMA, the individual extreme learning machine models, least squares support vector machines, and Artificial Neural Networks. Zhao et al. [14] proposed an ensemble of Deep learning models to propose the oil prices per barrel. The authors observe that the proposed model outperformed Feed-forward neural networks, support vector machines, Markov regime-switching, and random walk algorithm. Xiong et al. [15] hybridized ARIMA, EEMD, feedforward neural networks, support vector regression, and extreme learning machines to predict hog prices in China. It was observed that the hybrid model performed better than all the base models used. Ribeiro et al. [16] used bagging, boosting, and stacking to forecast agriculture commodities' prices one month ahead of time. They reported that the ensemble approach gave statistically significant gains and low prediction errors for their time series data. Wu et al. [17] proposed an ensemble model built on random forest, support vector regression, multilayer perceptron neural network and K-Nearest Neighbor regression in the first layer and Linear regression in the second layer for forecasting evotranspiration. Although the proposed model outperforms several benchmarks, the study does not incorporate Boosting trees and sequence models that could result in predictions with lower error rates.

Regression ensembles have been used to forecast data in the fields of energy, metal, livestock, and agriculture [18]. Pinto et al. [19] proposed three ensemble models — gradient boosted regression trees, random forests and a variant of gradient boosting for short term load forecasting. Real data from an office building was used to train a model that predicts electricity load of the building an hour ahead. It was observed that the variant of Adaboost outperforms the other models. However the authors did not compare these results with stacking which is known to generally outperform boosting and bagging algorithms. An ensemble model based on ANNs for forecasting electricity load and generation simultaneously was proposed by Jetcheva et al. [20]. The data was collected from commercial and industrial sites for a period of over 10 months. Khairalla et al. [21] proposed a stacking multi-learning ensemble for short term Global Oil consumption The proposed stacking model combines support vector regression, backpropagation neural network, and linear regression learners. It was observed that the proposed model outperformed traditional time series analysis models considered by the authors. The authors further conclude that the ensemble model is a very encouraging model for complex time series forecasting. A drawback of this study is that the popular regression models like Evolutionary Trees and Extreme Gradient boosting is not considered. These models could help to increase the performance of the proposed ensemble.

Al-Musaylh et al. [22] designed a hybrid ANN model that combines the output of 3 models - ANN, multivariate adaptive regression spline (MARS) and multiple linear regression (MLR). This model was used to predict short term electricity demand forecasting at 8 stations in Queensland, Australia. The hybrid ANN was outperformed the 3 base models as well as ARIMA but, it was not benchmarked against Sequence models and Ensemble models. Support vector regression (SVR) model was integrated with an improved version of empirical mode decomposition with adaptive noise (ICEEMDAN) tool and the weights were optimized using Particle Swarm optimization to form an hybrid model [23]. This two-phase model was used for multiple horizon electricity demand forecasting and was observed to outperform several benchmarks. However, SVR which is a machine learning model, does not perform well for larger datasets. Deep learning models are required to obtain scalable models. Moon et al. [24] proposed four deep neural models with different hyper-parameters and the predictions of these four models were combined using sliding window-based principal component regression (PCR). The model was observed to be highly accurate and stable for multi-step forecasting. However, PCR is not as effective as boosting trees when a non-linear combination of the predictions of the base models is required.

Bedi et al. [25] used LSTMs to develop a multi-input–multi-output model for electricity demand forecasting in Chandigarh, India [26].

Their approach was better than having multiple single regression models for multi-step forecasting as it takes into consideration the dependencies between the items in the output as well. This model was observed to outperform several state-of-art prediction models. However, the results obtained by this model are not compared with popular ensemble learning models for electricity demand forecasting. A combination of outlier detection, feature selection and an ensemble approach using K Nearest Neighbors and Naive Bayes classifier was proposed for load forecasting by Saleh et al. [27]. Although the proposed approach outperforms other load Forecasting strategies, a comparison with more advanced deep learning and machine learning models was not attempted. To deal with the problem arising due to the limited amount of data available, Ozer et al. [28] proposed a transfer learning approach to leverage the data from other places in the world. The model was evaluated on an actual energy system. However, the results have not been bench-marked against any recent works in load forecasting.

Some interesting solutions have been proposed to electricity consumption forecasting in Spain using the dataset that we consider in this paper. Jose et al. [29] used deep feed forward neural networks provided by H2O big data analysis framework to not only yield accurate predictions but enable the solution to be scalable. A grid search is proposed to estimate the correct hyper-parameters of the model. Torres et al. [30] proposed a neural networks model for forecasting and they used random search to tune multiple hyper-parameters involved with the model. The predictions obtained are then smoothed using the moving average method. It was observed that the random search for hyper-parameter tuning coupled with smoothing resulted in low prediction errors. The drawback of both these solutions is that they both considered historical windows of size 168. This requires more data to be stored in the database

Based on the existing research, it can be seen how ensemble models outperform traditional time series forecasting approaches, especially those that use a single model for forecasting. Divina et al. [10] proposed a novel ensemble model to forecast short-term electricity consumption in Spain. The proposed ensemble approach has not considered the latest developments in the field of time series forecasting and machine learning. This approach does not pre-process the dataset, while we use differencing to make the data stationary. Grid search is used to estimate the hyper-parameters, the drawback of which is that the search space is limited. However, we use random search to explore a larger search space with the same computational cost. Instead of shallow neural networks used in state-of-the-art approaches, we use Deep Neural Networks, in the form of ensemble models. We adopt LSTM in addition to the 3 base models in the first layer of the ensemble model.LSTM can model the dependencies between the items in the prediction horizon which cannot be done by the single-output regression models used in the current solution . We include XGB instead of GBM in the second layer of the ensemble-model to obtain a more scalable model when compared to state-of-the-art. The proposed approaches are benchmarked against state-of-the-art works [7,31,32].

3. Proposed methodology

3.1. Data preprocessing

For the experimental validation of our proposed work, we considered a dataset containing nearly 500,000 electricity consumption values collected at periodic intervals over the span of 9 years [10] Data has to be made stationary as that will remove the seasonality and trend in the data. Since most machine-learning and deep-learning models perform their best when the data is independent, making the data stationary can result in significantly better results. We adopted *Differencing* [31] to make the data stationary, as it involves transforming the time series into a new time series where values are the differences between adjacent values from the original dataset. Although the data is stationary according to multiple hypothesis tests, preprocessing using differencing allows us to lower the significance levels in these hypothesis tests. Differencing helps to stabilize the mean of time series data by removing the changes in the levels. Temporal structure might persist even after one level of differencing. Differencing might have to be applied multiple times till these temporal structures are completely removed. (see Figs. 1–3).

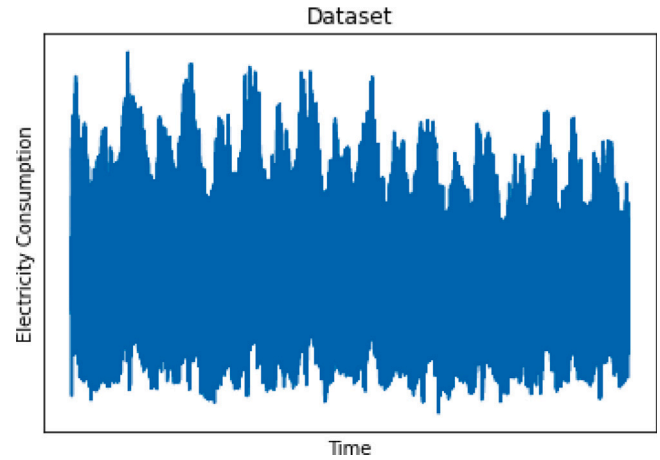

Fig. 1. Original dataset.

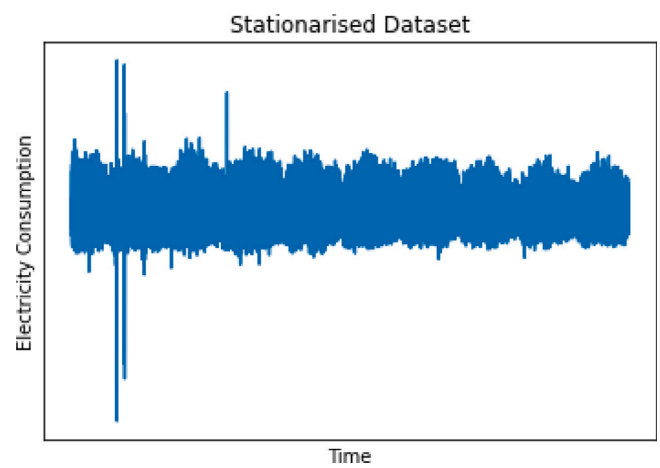

Fig. 2. Dataset after applying 1 level of differencing.

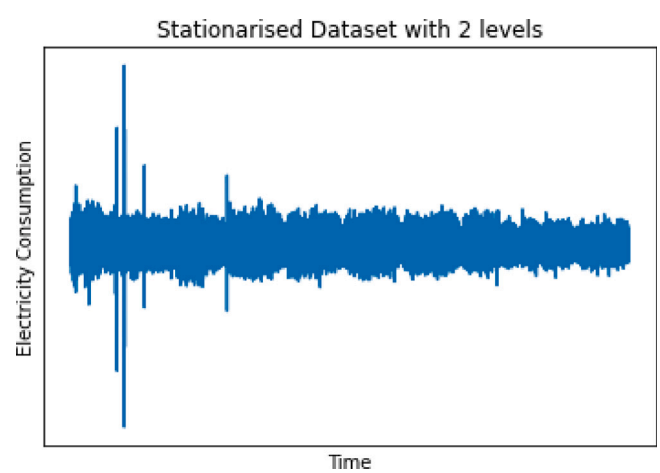

Fig. 3. Dataset after applying 2 levels of differencing.

Table 1
Stationarity of time series data results.

Test statistic	Level	Value
ADF statistic	Original dataset	−31.464335
	1-level of differencing	−68.238505
	2-levels of differencing	−43.828337
PP statistic	Original dataset	−25.632
	1-level of differencing	−262.442
	2-levels of differencing	−159.542

We used Augmented Dickey–Fuller (ADF) hypothesis test [32] and Phillips Perron test [33] to test the stationarity of the dataset and to determine the order of differencing. The ADF test essentially is used to examine the null hypothesis and determine if a unit root is present in the time series data. A Unit root is a characteristic of a time series that makes it non-stationary. The more negative the test statistic is, the stronger is the rejection of this null hypothesis. The Phillips Perron test is similar to the ADF test, but with the additional advantage of being non-parametric, that is, it does not require to select the level of serial correlation as in ADF. From Table 1, it is evident that the lowest (or largest absolute) ADF statistic and PP statistic occurs for the dataset when 1 level of differencing is performed. This indicates that the rejection of the null hypothesis is the strongest when the order of differencing is 1 as compared to when the order is 0 or 2.

We use two different approaches for forecasting. In our first approach, we set the size of the historical data to 96 (16 h), while in the second, the size of the historical data is set to 168 (28 h). In both the cases, the prediction horizon is set to 24 (4 h). This means that in our first approach, data from the last 16 h was used to predict the consumption values for the next 4 h, and in our second approach, data from the last 28 h to predict the same. The window is slid by 24 values each time and the total number of records is 20,739. 70% of this data is used for training and the remaining 30% is used for testing.

3.2. Model ensembling

The proposed prediction model is an ensemble model, and we experimented with two stacking ensemble models. The weak learners in both the models are the same – Long short term memory (LSTM), Evolutionary trees (EvTree), Random Forests (RF) and Neural Networks (NN). Ensemble-1 uses Gradient Boosting Model (GBM), whereas, Ensemble-2 uses Extreme Gradient Boosting (XGB) to combine the predictions of the 4 weak learners using a stacking approach. Except for LSTM, all the models use the multi-model approach. This means that 24 different models are used to predict each of the 24 periods in the prediction horizon.. The hyperparameters of the models are determined using random search over the parameters that were used in other similar works. In the section, we give a brief description of each model and the optimal hyper-parameters for each of these models.

3.2.1. Random forests classifier

Random forests are a combination of multiple decision tree predictors, whose predictions are combined by bagging [34]. An average of the predictions of each tree is presented as the output of the model. Each tree is trained on a random sample from the dataset that is obtained through bootstrapping. Random Forests do not use all the features from the input. Each tree is trained using a random sample of the features. This results in a low correlation between the trees. These reasons make Random Forests less prone to overfitting. The hyper-parameters of the Random Forests model are estimated using a Random search strategy. The hyper-parameters that are used in similar research are explored in the random search. Table 2 contains the best hyper-parameters obtained after random search.

Table 2
Optimal hyperparameter values derived for Random Forests.

Hyperparameter	Value
n_estimators	200
max_leaf_nodes	150
min_samples_leaf	1
min_samples_split	2

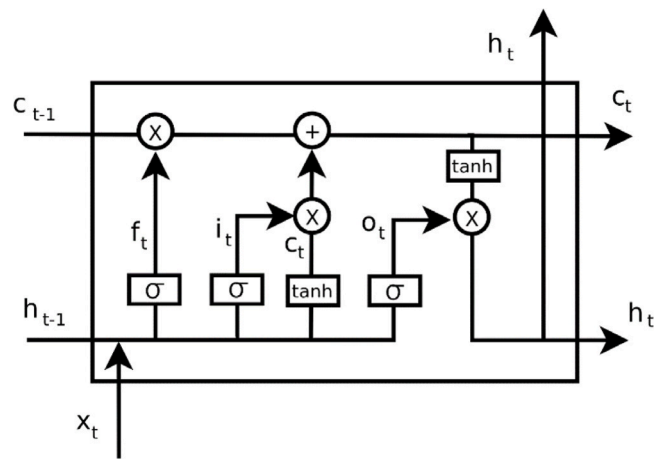

Fig. 4. Architecture of an LSTM Cell [35].

3.2.2. Long Short Term Memory (LSTM)

LSTM is a type of Recurrent Neural Network (RNN) that performs well on time-series data [35]. The difference between a RNN and a traditional machine-learning model or a neural network is that it has a concept of memory. The output of any layer depends on the output of the previous layers as well. Each LSTM cell has a cell state as well as a hidden state. LSTMs have 3 gates: input, output and forget, and the cell states are controlled by these gates (see Fig. 4). The forget gate helps to identify how much data from x_t and $h_t - 1$ is irrelevant and how much should be forgotten. The input gate decides how much the cell state has to be updated and the output gate determines what the next hidden state should be. Here x represents the input, h represents the hidden state, and t represents the time step. LSTM does not suffer from the vanishing gradient problem like Vanilla RNN [36], which makes it very effective in remembering long-term dependencies in sequences. It also has the ability to learn the context required to make predictions in time series forecasting problems.

The LSTM model used in our work consists of 4 LSTM layers with 250, 150, 100 and 50 units respectively, followed with an output layer with 24 nodes. The objective function to be minimized is Mean Squared Error (MSE), and Adam optimizer with a learning rate of 0.01 is used. The input has to be normalized before feeding it to the LSTM model, for which the MinMax scaler was used [37]. The min and max parameters are set to −1 and +1 respectively and all the values are transformed into the range [−1,1]. This is done in order to normalize the data and bring all the values in the same range. Backpropagation in LSTM tends to be more stable and faster when the data is scaled. The output of the LSTM model must be rescaled to the original scale. This is the only multi-input, multi-output model among the ensemble models proposed in this paper. The entire prediction horizon is forecasted using a single LSTM model. The other models assume that all the items in the prediction horizon are independent, however, LSTM models the dependencies between these items as well.

3.2.3. Deep Neural Networks (DNN)

An Artificial Neural Network (ANN) has 3 layers — input, hidden and output [38]. The number of hidden layers determines whether an ANN is deep or shallow, and an ANN with 2 or more hidden layers is

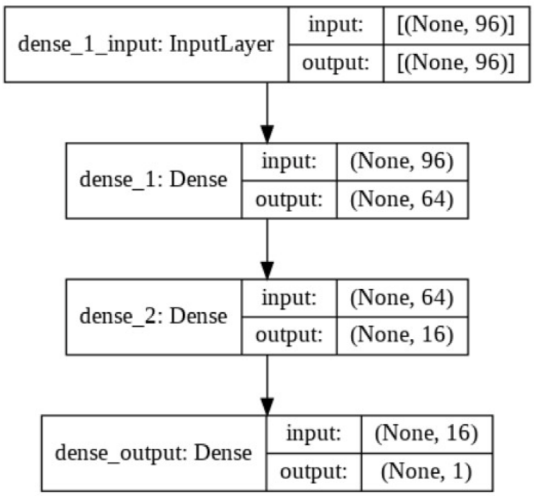

Fig. 5. Architecture of the DNN model.

called a Deep Neural Network [39]. The activation function in each hidden layer introduces non-linearity. When the number of hidden layers is increased the performance of the network increases because more complex non-linear relationships are learnt. However, very deep neural networks can be overfitting and fail to generalize well [26]. Hence, it is important to determine the right number of hidden layers. Also the number of nodes in each hidden layer has to be determined by experimentation. We consider a 4 layer network, with 1 input, 2 hidden layers and 1 output layer (Fig. 5). The 96 and 168 values in the historical data are fed to the input layer of the neural network. The 2 hidden layers have 64 and 16 nodes respectively, which are followed by the output layer that has 1 node. ReLu activation function is used at the hidden layers and linear activation function is used at the output layer. Mean squared error objective function and Adam optimizer with a learning rate of 0.1 is used.

3.2.4. Evolutionary trees

Evolutionary Algorithms (EAs) for Decision Trees or in our case for regression are genetic algorithms in which each genome of the population is a regression tree and uses evolutionary techniques such as selection, crossover and mutation [40]. These algorithms have certain drawbacks since the search space can be huge; sometimes these methods give sub-optimal solutions and these methods tend to produce complex tree structure that overfit on the training data and such does not generalize. On the other hand EAs have robust search capabilities and can be used for huge search spaces. The use of stochastic operators, such as mutation, allows EAs to escape from local optima.

To implement the EATree model, firstly, a set of regression trees are randomly initialized, to produce a population of random trees. Each candidate or tree in the population is evaluated based on a fitness function. The fitness function is based on two criteria i.e the accuracy of the regression tree on the training set (higher the better) and the size of the tree which is usually the depth of the tree (smaller the better). The next step is selection of the fittest candidates is done. The selection process is based on the fitness of the tree and the trees that are fitter have higher chances of getting selected for the crossover process (roulette selection). Now, the next step of crossover of the selected candidates is performed in order to obtain the next generation of candidates. Here, 2 candidates or parents are selected and then combined at some random node, replacing the subtree of one of the parents to produce a child. This process is continued until the required population size is reached. The last step is mutation. Mutation is done in order to increase genetic diversity and to escape from local minima if stuck. During this phase, some offspring are selected based on the mutation rate. A node is chosen at random for each selected tree and changes the attribute and the split at that node. These steps are repeated until the number of generations has crossed a threshold or the fitness of the trees have converged i.e no improvement between generations. We use the same hyperparameters as Divina et al. [10].

Table 3
Optimal Hyperparameter values for XGB and GBM.

Hyperparameter	Value
learning_rate	0.09
n_estimators	170
max_depth	4
min_child_samples	20
min_child_weight	0.001
num_leaves	31

3.2.5. Gradient Boosting Model (GBM)

GBM [41] is an ensemble technique where regression trees are used as weak learners, which can be defined as stage-wise additive models. This method adds new learners at each stage, sequentially using a gradient descent approach. Weak learners are added till a given number of trees have been built or if there is no improvement in the performance of the model. When a new weak learner is added, the existing weak learners are frozen and left unchanged. A mean squared error loss function is minimized in a regression problem. GBMs are prone to overfitting and regularization can be used to reduce overfitting. Once all the trees have been constructed, the final predictions are obtained by boosting technique. A weighted average of the predictions from all trees is presented as the final prediction of the model.

3.2.6. Extreme Gradient Boosting (XGB)

XGB, like GBM, uses gradient descent to build an ensemble of trees [42]. A major difference between XGB and GBM is that XGB uses a more regularized model formation to reduce overfitting. XGB implements Dropout regularization for regression trees (DART) [43]. This enables XGB to generalize well and have a low bias and low variance. Another advantage of XGB over GBM is that XGB is scalable. XGB makes use of sparse matrices with sparsity aware algorithms. XGB also uses efficient data structures for better processor cache utilization, and has support for multi-core parallel processing. Hence, when using a GPU accelerator, the computation is speeded up significantly. The amount of time to train an GBM model is at least 10 times that of an XGB model. Similar to the other models used in our work, the performance of an XGB model depends on the hyper-parameters. A random search over the parameters that are used in literature for similar applications, is performed. We use the same parameters even for the GBM model (see Table 3).

4. Experimental results and analysis

For the experimental validation of the proposed approaches, five regression measures were used — Mean relative error (MRE), coefficient of determination R^2, the mean absolute error (MAE), the root mean squared error (RMSE) and the symmetric mean absolute percentage error (SMAPE). MRE is the average of the ratio of the absolute difference between the predicted value and the actual value to the actual value over all the predictions and is given by Eq. (1). MAE is the mean of the absolute difference between predicted value and actual value over all the predictions and is computed as per Eq. (2). SMAPE is an accuracy measure based on percentage errors (Eq. (3)). It is better protected against outliers as compared to the MRE and MAE. R^2 indicates the percentage of error reduced by the model when compared to a naive model that simply outputs the mean of the actual values in the dataset, given by Eq. (4). RMSE is the square root of the average of squared errors, and is directly proportional to the squared errors, hence larger

squared errors result in larger RMSE as well. RMSE, unlike SMAPE, is sensitive to outliers and is computed as per Eq. (5). In Eqs. (1) through (5), n corresponds to the number of predictions, e_t is the difference between the actual value and the value predicted by the model, $\bar{y}_t$ is the value predicted by the model and y_t is the actual value.

$$MRE = \frac{1}{n}\sum_{t=1}^{n}\frac{|e_t|}{y_t} \tag{1}$$

$$MAE = \frac{1}{n}\sum_{t=1}^{n}|e_t| \tag{2}$$

$$sMAPE = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{2e_t}{y_t+\hat{y}_t}\right| \tag{3}$$

$$R^2 = 1-\sum_{t=1}^{n}\frac{e_t^2}{|y_t+\bar{y}_t|} \tag{4}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}e_t^2} \tag{5}$$

For historical window of size 96, we compare our results with that of the ENSEMBLE model proposed by Divina et al. [10]. For the historical window of size 168, we benchmark our results with three state-of-the-art works — ENSEMBLE model [10], Feed forward neural networks (FNN-H20) approach of Jose et al. [29] and Torres et al.'s DNN-Smoothing approach [30]. Also, we compare the performance with traditional approaches like Linear regression, ARMA, ARIMA and other deep learning based ensemble approaches. Our objective is to show the wide scope of our proposed approach, and to demonstrate its superiority over other existing approaches as well as traditional approaches.

Table 4 represents the results of the ensemble model and the base layer models on the original dataset for historical window of size 96, i.e without applying the preprocessing step. Tables 5 and 6 present the observations w.r.t performance of various models noted for the prediction horizon of 24 and historical window of 96 & 168 respectively. Our proposed models, *Ensemble-1* and *Ensemble-2* performed equally well, while outperforming all other existing works and base-level models. The results of the existing works are presented without preprocessing data similar to how it was presented in the works. It was observed that *Ensemble-2* had a much lower training time when it was accelerated with GPU as compared to *Ensemble-1*. The results of both the proposed ensemble models were much better when compared to ENSEMBLE [10], FNN-H20 [29] and DNN-smoothing [30]. It can also be observed that there is a reduction of approximately 17.4% in the root mean squared error of the predictions of the ensemble model when differencing is applied. The predictions from all the models are better except for Evtrees. Three out of the four base models used were the same as the ensemble approach proposed by Divinia et al. [10], however, the preprocessing as well as better hyper-parameter tuning by random search shows the results of these base models are much better as well. An interesting observation is that the prediction errors for the proposed approach for historical window of size 96 is better than the results of the ensemble model for historical window of size 168. Although our work is an extension of Divina et al. [10], the novel improvements incorporated in the proposed methodology help obtain significantly better results.

Table 7 presents the MAE of the models for each of the prediction horizon of 24 and historical window of 96 & 168. Fig. 6, 7 and 8 demonstrate the trends in prediction errors for each h in the prediction horizon for $w = 96$. Similarly, Fig. 9, 10 and 11 show the trend in prediction errors for $w = 168$. The values of MAE, RMSE, MRE and sMAPE show an increasing trend while R^2 shows a decreasing trend. The ensemble models predict 24 values at once, as it is easier to predict the first few values of the precision horizon as compared to the last ones. This is because the dependency between the historical data and the prediction horizon decreases as h increases. Hence, the predictions

Table 4
Comparative evaluation for Historical Window of 96 without differencing.

Model	MRE	R^2	MAE	RMSE	sMAPE
Ensemble-1 *(Proposed)*	0.44	0.96	522.64	748.26	0.02
Ensemble-2 *(Proposed)*	**0.44**	**0.97**	**521.66**	**746.94**	**0.02**
Random Forests	0.67	0.94	768.15	1032.84	0.03
Evtree	1.02	0.91	942.47	1277.09	0.03
NN	0.48	0.96	592.07	792.68	0.02
LSTM	1.16	0.85	1372.27	1800.14	0.04

Table 5
Comparative evaluation for Historical Window of 96 with differencing.

Model	MRE	R^2	MAE	RMSE	sMAPE
ENSEMBLE [10]	2.57	0.95	714.16	969.66	0.02
Ensemble-1 *(Proposed)*	0.39	0.98	453.49	637.87	0.02
Ensemble-2 *(Proposed)*	**0.39**	**0.98**	**453.42**	**636.10**	**0.02**
Random Forests	0.61	0.95	696.78	982.79	0.03
Evtree	1.04	0.87	1184.63	1661.15	0.04
NN	0.41	0.98	467.03	649.98	0.02
LSTM	1.16	0.86	1315.47	1735.24	0.05
ARMA	4.66	0.86	1264.58	1619.73	0.04
ARIMA	6.79	0.81	1858.88	2282.80	0.09

Table 6
Comparative evaluation for Historical Window of 168.

Model	MAE	RMSE
ENSEMBLE [10]	513.50	745.31
FNN-H20 [29]	440.64	587.46
DNN-Smoothing [30]	369.19	501.14
Ensemble-1 *(Proposed)*	309.06	421.06
Ensemble-2 *(Proposed)*	**308.53**	**421.03**
Random Forests	557.65	788.03
Evtree	1184.51	1660.84
NN	332.08	444.46
LSTM	855.00	1194.94
ARMA	660.20	889.18
ARIMA	1836.62	2332.30

are better for the first few values in the historical window as compared to the predictions towards the end of the prediction window. The lines for XGB and GBM almost overlap in all the graphs which is because of the closeness in their predictions.

As per our observations. the Deep neural network was the best performing base model. LSTM did not perform as well as Deep neural networks because the seasonality or trend in the data is reduced by the preprocessing step. Hence, the data is largely stationary and independent. However, there could be a few cases in the dataset, where the seasonality and trends are not completely eliminated. In such cases, a greater priority is assigned to the predictions from LSTM. Hence, even though LSTM and EVTrees do not perform as well as the other base models, eliminating them will reduce the accuracy of the final predictions. Figs. 12 and 13 give a graphical representation of the relative performance of all 5 models used in this paper. It can seen that, *Ensemble-1* and *Ensemble-2* perform almost the same, while both the proposed ensemble approaches outperform state-of-the-art and individual base models used for comparison.

Discussion. In the proposed approach, we incorporate several improvements that contribute to their superior performance. Firstly, we perform differencing to make the data stationary as that will remove any form of dependencies in it. Although the data is stationary according to the multiple hypothesis tests performed (See Table 1), preprocessing using differencing allows us to lower the significance levels in these hypothesis tests. Since most machine-learning and deep-learning models perform their best when the data is independent, making the data stationary can result in significantly better results, as is evident in this case. Also, we consider Deep Neural Networks in place of the shallow neural networks in contrast to state-of-the-art approaches. To

Table 7
MAE for each prediction horizon value (h).

Models/h	w = 96						w = 168					
	RF	Evtree	NN	LSTM	Ensemble-1	Ensemble-2	RF	Evtree	NN	LSTM	Ensemble-1	Ensemble-2
1	**176.82**	347.15	160.49	709.72	151.89	151.45	175.27	347.15	172.27	507.1	145.85	145.28
2	**267.86**	472.29	204.12	630.06	197.19	196.87	263.82	472.34	172.56	439.02	173.62	172.98
3	**338.63**	695.56	232.08	691.85	240.63	240.28	319.54	695.61	191.25	480.43	190.34	190.11
4	381.8	717.89	263.35	778.54	260.52	260.37	344.95	717.68	212.94	499.57	201.44	201.39
5	431.06	842.7	304.11	878.04	296.29	295.45	392.86	842.77	251.58	533.8	220.3	219.86
6	469.2	950.27	338.98	840.98	326.59	328.3	422.65	950.31	288.31	789.14	231.42	229
7	511.96	948.46	388.77	1032.26	362.32	359.19	460.27	948.51	310.53	827.82	255.45	252.64
8	573.69	1094.79	418.43	1193.16	398.33	397.38	483.64	1094.76	272.42	746.26	256.86	258.42
9	606.68	1072.53	431.05	1279.14	418.95	417.65	498.63	1072.56	336.21	853.44	282.39	282.4
10	660.48	1199.61	455.87	1189.28	430.08	430.28	535.95	1199.06	303.32	763.41	284.97	284.35
11	729.01	1352.65	464.37	1305.91	454.68	450.04	581.68	1352.78	308.39	1176.54	299.46	299.74
12	773.47	1247.62	450.3	1516.87	443.21	442.58	600.22	1246.63	311.43	920.15	305.1	304.78
13	826.92	1247.44	484.5	1219.89	466.18	462.05	612.24	1247.64	351.45	892.81	325.26	325.21
14	848.14	1305.73	465.27	1412.34	462.53	463.08	598.89	1305.94	331.51	846.41	316.44	318.2
15	826.33	1369.78	486.33	1582.73	480.91	484.2	603.57	1369.94	378.26	975.43	343.38	343.72
16	828.23	1431.29	516.39	1597.35	503.26	505.44	631	1431.49	362.4	840.74	347.18	345.96
17	848.42	1459.37	530.28	152.74	525.68	527.11	655.1	1458.78	382.61	982.38	351.64	351.49
18	862.2	1428.82	580.6	1675.36	543.5	545.41	654.12	1428.00	367.96	1018.35	365.76	365.36
19	881.25	1459.11	625.2	1623.01	575.32	575.44	665.44	1458.33	400.66	978.75	374.24	374.13
20	914.93	1449.25	598.05	1675.34	595.22	599.74	700.95	1449.30	448.75	1180.52	392	390.27
21	943.29	1490.87	680.33	1659.73	659.77	662.95	775.06	1491.01	456.36	955.30	429.79	428.2
22	975.3	1588.33	703.97	1536.96	683.1	685.98	767.47	1587.93	431.63	1026.78	425.19	422.87
23	1000.43	1634.70	703.01	1875.35	698.56	697.05	811.81	1634.86	456.32	1149.51	444.25	444.42
24	1046.57	1624.99	722.89	2314.63	707.31	705.59	828.55	1624.92	470.69	1136.35	455.1	453.87

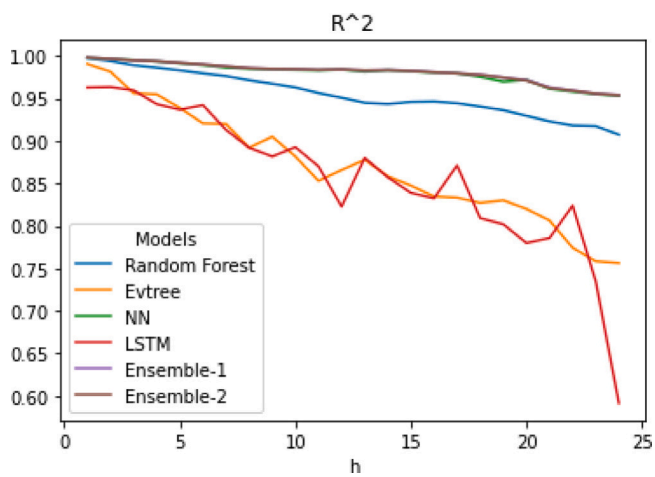

Fig. 6. Comparison of R^2 of each model for w = 96.

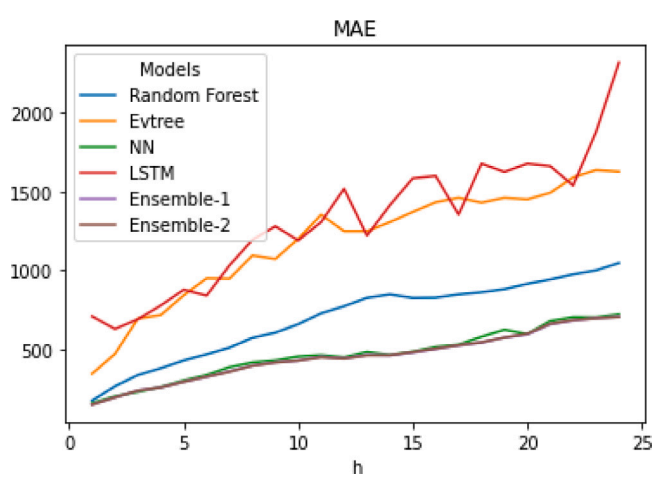

Fig. 7. Comparison of MAE of each model for w = 96.

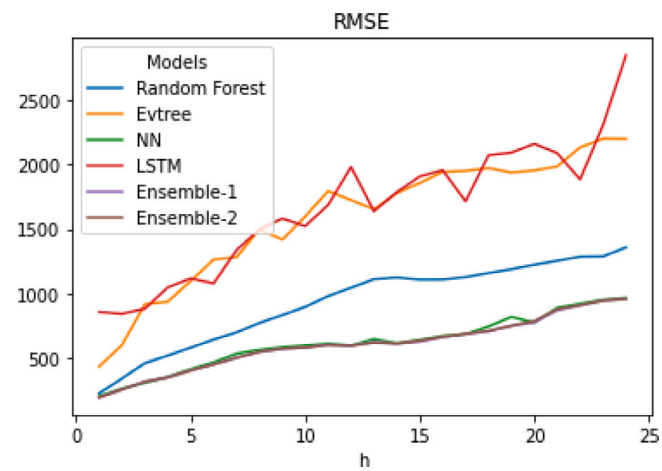

Fig. 8. Comparison of RMSE of each model for w = 96.

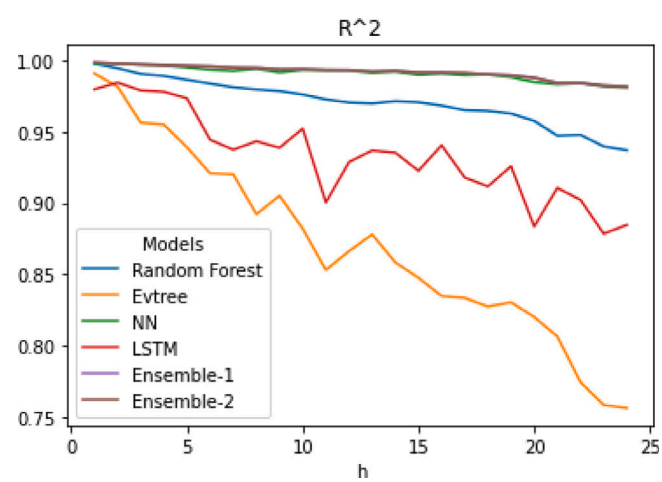

Fig. 9. Comparison of R^2 of each model for w = 168.

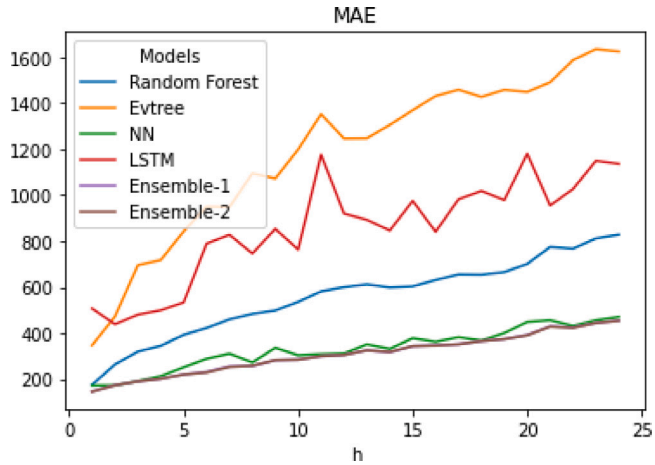

Fig. 10. Comparison of MAE of each model for w = 168.

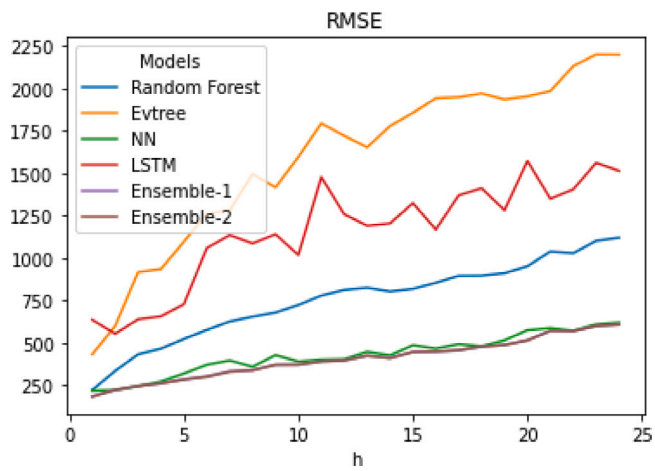

Fig. 11. Comparison of RMSE of each model for w = 168.

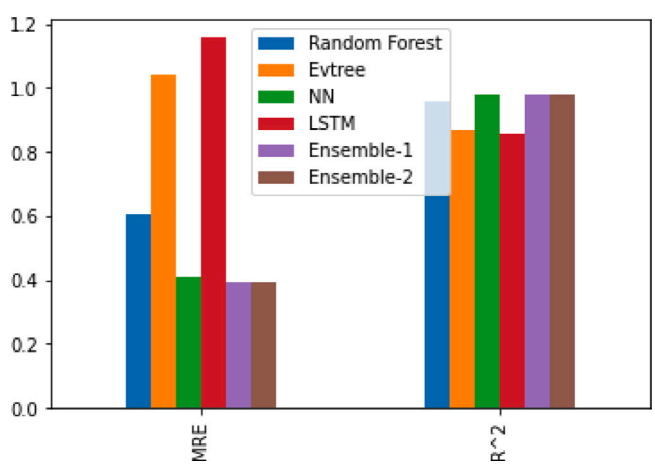

Fig. 12. Comparison of MRE and R^2 of each model for w = 96.

obtain the same accuracy, Deep Neural Networks are more efficient in terms of computation and number of parameters, such that at each hidden layer of the neural network, a more abstract representation of the input is learned. This makes it more effective in modeling non-linear dependencies between the input and output compared to a shallow neural network. The hyperparameters of the models used are

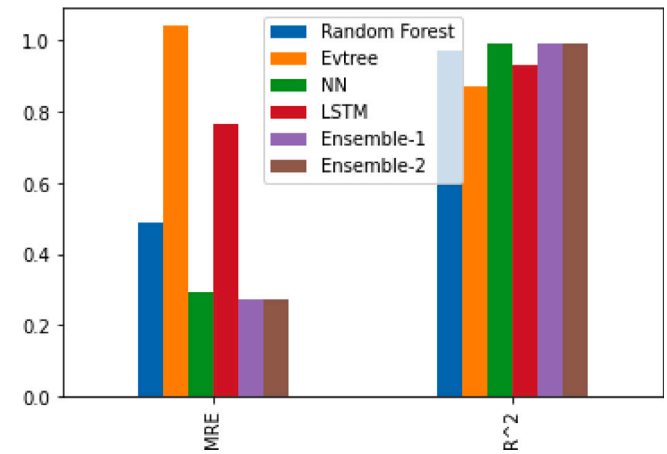

Fig. 13. Comparison of MRE and mean R^2 of each model for w = 168.

Table 8
Results of Diebold–Mariano test.

h parameter	Model A	Model B	DM Statistic	p-value
h = 96	Proposed model 1	ENSEMBLE [10]	−3.0554	0.005
h = 168	Proposed model 2	ENSEMBLE [10]	−2.7636	0.011

estimated by a Random Search. Divina *et al.* [10] use a grid search over the parameters for similar applications. The search space using this approach is limited, however, using Random search, a larger search space can be explored with the same computational cost. Even a small change in the hyperparameters could affect the performance of the models, hence it is essential to precisely estimate the hyper-parameters.

We also introduce LSTM as the fourth base model in the first layer of the ensemble model. LSTMs can learn the ordered dependencies between items in a sequence, making them very effective for time-series forecasting.LSTMs is the only multi-input-multi-output model being used in this paper. Hence, it is responsible for modeling the dependencies between the items in the prediction horizon Extreme gradient boosting (XGB) is used in the second layer of the ensemble. XGB is a variant of the gradient boosting method, which allows for parallel computing. Hence with GPU, XGB can significantly reduce the computation time required for training and predicting. This makes our approach much faster and scalable as compared to the current solutions. The proposed model considers a smaller historical window but still gives more accurate predictions than the existing state-of-the-art solutions. This implies that the database can store a lesser amount of data, however, using a larger historical window, will improve the accuracy of the predictions. We leave it to the user to decide which historical window size to use based on the memory available and the compromise in accuracy that he/she is comfortable with.

We also use the Diebold–Mariano test [44] to prove that the forecasts of our proposed ensemble model are significantly different when compared to state-of-the-art works. Mean squared errors are used as the loss function. We defined the null and alternate hypothesis as follows:

H_0 : The forecast errors are not significantly different.

H_1 : The forecast errors are significantly different.

The two-tailed test with a significance level set at $\alpha = 0.05$ which corresponds to $z_{crit} = -1.96\& + 1.96$ is used for the experiments. As can be observed from the results of this test tabulated in Table 8, the DM test statistic for prediction horizon of 96 is $-3.0554 < -1.96$ and the DM test statistic for prediction horizon of 168 is $-2.7636 < -1.96$. Therefore, the forecast errors for our model is significantly different from ENSEMBLE [10].

The proposed models are composed of an ensemble of 5 machine learning and deep learning models. Hence, the training as well as prediction time for the entire ensemble model is higher when compared

to a simpler solution that contains just a single model. The same model cannot be applied for other real-time application that has to process multiple inputs at the same time. Although the accuracy of the output will still be high, the waiting time for the users of the application will also be high making it inconvenient to use. In such cases a trade off has to be made between the convenience of the application and the accuracy of predictions.

5. Conclusion and future work

In this paper, a stacking ensemble model for short-term energy consumption forecasting was discussed. We adopt differencing to remove seasonality in the data, to make the data stationary. Deep neural models are adopted in place of the shallow neural networks, for better modeling non-linear dependencies between the input and output compared to a shallow neural network. The hyperparameters of the models are estimated by a Random Search, so that a larger search space can be explored with the same computational cost. An LSTM and XGBoost ensemble is proposed, for much faster and scalable predictions, when compared to the state-of-the-art. Experimental results showed that the proposed Ensemble-2 model performed best in comparison to all state-of-the-art and base models. Although ensemble-1 and ensemble-2 have very similar performance scores, ensemble-2 is much faster and scalable. The XGB and GBM model accurately combine the predictions of the weak learners by assigning different priorities for each base layer model depending on the input. The results of the proposed solutions are significantly better in comparison with the solutions and also traditional approaches to time series forecasting. Predicting electricity consumption accurately is crucial for garnering insights and potential trends into energy consumption for effective resource management. Due to the linearity/non-linearity in usage patterns, electricity consumption prediction is challenging and cannot be adequately solved by using a single model. The proposed ensemble models combine the hypothesis of multiple models which allows for a closer approximation of the true hypothesis and overcomes the limitation of using single regression models. We also perform multi-step forecasting so the energy consumption values for multiple time steps are predicted at once.

The proposed approach can be applied to any time series data, as after performing preprocessing to make the data stationary as per the dataset requirement, all time series problems are similar. Hence, to a certain extent, the proposed approach can be generalized for any dataset. Also, a hierarchical ensemble approach could be adopted for improving future experimental analysis. One could also develop different ensemble models for each value in the prediction window. We used the same hyper-parameters are used to predict all the 24 values in the historical window. A rolling forecast model could help to smoothen the predictions. In the dataset, there are a few outliers where the electricity consumption values are abnormally high or low. By using a rolling forecast to smoothen it, the prediction errors could be lowered. A moving average approach could also be used for smoothening the predictions. As part of future work, we also plan to extend the models for predictions in different domains, like agricultural commodities price and stock price prediction.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] R. Haque, T. Jamal, M.N.I. Maruf, S. Ferdous, S.F.H. Priya, Smart management of PHEV and renewable energy sources for grid peak demand energy supply, in: 2015 International Conference on Electrical Engineering and Information Communication Technology (ICEEICT), IEEE, 2015, pp. 1–6.

[2] M. Zekić-Sušac, S. Mitrović, A. Has, Machine learning based system for managing energy efficiency of public sector as an approach towards smart cities, Int. J. Inform. Manage. 58 (2021) 102074.

[3] M. Manic, D. Wijayasekara, K. Amarasinghe, J.J. Rodriguez-Andina, Building energy management systems: The age of intelligent and adaptive buildings, IEEE Ind. Electron. Mag. 10 (1) (2016) 25–39.

[4] U. Thissen, R. Van Brakel, A. De Weijer, W. Melssen, L. Buydens, Using support vector machines for time series prediction, Chemometr. Intell. Lab. Syst. 69 (1–2) (2003) 35–49.

[5] B. Freisleben, Stock market prediction with backpropagation networks, in: International Conference on Industrial, Engineering and Other Applications of Applied Intelligent Systems, Springer, 1992, pp. 451–460.

[6] S. Siami-Namini, N. Tavakoli, A.S. Namin, A comparison of ARIMA and LSTM in forecasting time series, in: 2018 17th IEEE International Conference on Machine Learning and Applications (ICMLA), IEEE, 2018, pp. 1394–1401.

[7] D. Opitz, R. Maclin, Popular ensemble methods: An empirical study, J. Artif. Intell. Res. 11 (1999) 169–198.

[8] T.G. Dietterich, Ensemble methods in machine learning, in: International Workshop on Multiple Classifier Systems, Springer, 2000, pp. 1–15.

[9] S. Džeroski, B. Ženko, Is combining classifiers with stacking better than selecting the best one? Mach. Learn. 54 (3) (2004) 255–273.

[10] F. Divina, A. Gilson, F. Goméz-Vela, M. García Torres, J.F. Torres, Stacking ensemble learning for short-term electricity consumption forecasting, Energies 11 (4) (2018) 949.

[11] L.A. Gabralla, H. Mahersia, A. Abraham, Ensemble neurocomputing based oil price prediction, in: Afro-European Conference for Industrial Advancement, Springer, 2015, pp. 293–302.

[12] C. Pierdzioch, M. Risse, S. Rohloff, Forecasting gold-price fluctuations: a real-time boosting approach, Appl. Econ. Lett. 22 (1) (2015) 46–50.

[13] L. Yu, W. Dai, L. Tang, A novel decomposition ensemble model with extended extreme learning machine for crude oil price forecasting, Eng. Appl. Artif. Intell. 47 (2016) 110–121.

[14] Y. Zhao, J. Li, L. Yu, A deep learning ensemble approach for crude oil price forecasting, Energy Econ. 66 (2017) 9–16.

[15] X. Tao, L. Chongguang, B. Yukun, An improved EEMD-based hybrid approach for the short-term forecasting of hog price in China, Agric. Econ. 63 (3) (2017) 136–148.

[16] M.H.D.M. Ribeiro, L. dos Santos Coelho, Ensemble approach based on bagging, boosting and stacking for short-term prediction in agribusiness time series, Appl. Soft Comput. 86 (2020) 105837.

[17] T. Wu, W. Zhang, X. Jiao, W. Guo, Y.A. Hamoud, Evaluation of stacking and blending ensemble learning methods for estimating daily reference evapotranspiration, Comput. Electron. Agric. 184 (2021) 106039.

[18] F.J. Fabozzi, Handbook of Finance, Financial Markets and Instruments, Vol. 1, John Wiley & Sons, 2008.

[19] T. Pinto, I. Praça, Z. Vale, J. Silva, Ensemble learning for electricity consumption forecasting in office buildings, Neurocomputing 423 (2021) 747–755.

[20] J.G. Jetcheva, M. Majidpour, W.-P. Chen, Neural network model ensembles for building-level electricity load forecasts, Energy Build. 84 (2014) 214–223.

[21] M.A. Khairalla, X. Ning, N.T. Al-Jallad, M.O. El-Faroug, Short-term forecasting for energy consumption through stacking heterogeneous ensemble learning model, Energies 11 (6) (2018) 1605.

[22] M.S. Al-Musaylh, R.C. Deo, J.F. Adamowski, Y. Li, Short-term electricity demand forecasting with MARS, SVR and ARIMA models using aggregated demand data in Queensland, Australia, Adv. Eng. Inform. 35 (2018) 1–16.

[23] M.S. Al-Musaylh, R.C. Deo, Y. Li, J.F. Adamowski, Two-phase particle swarm optimized-support vector regression hybrid model integrated with improved empirical mode decomposition with adaptive noise for multiple-horizon electricity demand forecasting, Appl. Energy 217 (2018) 422–439.

[24] J. Moon, S. Jung, J. Rew, S. Rho, E. Hwang, Combination of short-term load forecasting models based on a stacking ensemble approach, Energy Build. 216 (2020) 109921.

[25] J. Bedi, D. Toshniwal, Deep learning framework to forecast electricity demand, Appl. Energy 238 (2019) 1312–1326, http://dx.doi.org/10.1016/j.apenergy.2019.01.113, URL: https://www.sciencedirect.com/science/article/pii/S0306261919301217.

[26] F. Panchal, M. Panchal, Approximating number of hidden layer neurons in multiple hidden layer BPNN architecture, Int. J. Comput. Sci. Mobile Comput. 3 (11) (2014) 455–464.

[27] A.I. Saleh, A.H. Rabie, K.M. Abo-Al-Ez, A data mining based load forecasting strategy for smart electrical grids, Adv. Eng. Inform. 30 (3) (2016) 422–448.

[28] I. Ozer, S.B. Efe, H. Ozbay, A combined deep learning application for short term load forecasting, Alexand. Eng. J. 60 (4) (2021) 3807–3818.

[29] J.F. Torres, A. Galicia, A. Troncoso, F. Martínez-Álvarez, A scalable approach based on deep learning for big data time series forecasting, Integr. Comput.-Aided Eng. 25 (4) (2018) 335–348.

[30] J.F. Torres, D. Gutiérrez-Avilés, A. Troncoso, F. Martínez-Álvarez, Random hyper-parameter search-based deep neural network for power consumption forecasting, in: International Work-Conference on Artificial Neural Networks, Springer, 2019, pp. 259–269.

[31] R. Adhikari, R.K. Agrawal, An introductory study on time series modeling and forecasting, 2013, arXiv preprint arXiv:1302.6613.
[32] Y.-W. Cheung, K.S. Lai, Lag order and critical values of the augmented Dickey–Fuller test, J. Bus. Econ. Stat. 13 (3) (1995) 277–280.
[33] J. Breitung, P.H. Franses, On Phillips–Perron-type tests for seasonal unit roots, Econom. Theory 14 (2) (1998) 200–221.
[34] L. Breiman, Random forests, Mach. Learn. 45 (1) (2001) 5–32.
[35] S. Hochreiter, J. Schmidhuber, Long short-term memory, Neural Comput. 9 (8) (1997) 1735–1780.
[36] R. Pascanu, T. Mikolov, Y. Bengio, On the difficulty of training recurrent neural networks, in: International Conference on Machine Learning, PMLR, 2013, pp. 1310–1318.
[37] I.S. Amiri, O.A. Akanbi, E. Fazeldehkordi, A machine-learning approach to phishing detection and defense, Syngress, 2014.
[38] R.E. Uhrig, Introduction to artificial neural networks, in: Proceedings of IECON'95-21st Annual Conference on IEEE Industrial Electronics, 1, IEEE, 1995, pp. 33–37.
[39] T.W. Gentry, B.M. Wiliamowski, L.R. Weatherford, A comparison of traditional forecasting techniques and neural networks, Intell. Eng. Syst. Artif. Neural Netw. 5 (1995) 765–770.
[40] R.C. Barros, M.P. Basgalupp, A.C. De Carvalho, A.A. Freitas, A survey of evolutionary algorithms for decision-tree induction, IEEE Trans. Syst. Man Cybern Part C 42 (3) (2011) 291–312.
[41] J.H. Friedman, Greedy function approximation: a gradient boosting machine, Annals Stat. (2001) 1189–1232.
[42] T. Chen, C. Guestrin, Xgboost: A scalable tree boosting system, in: Proceedings of the 22nd Acm sigkdd international conference on knowledge discovery and data mining, 2016, pp. 785–794.
[43] R.K. Vinayak, R. Gilad-Bachrach, Dart: Dropouts meet multiple additive regression trees, in: Artificial Intelligence and Statistics, PMLR, 2015, pp. 489–497.
[44] D. Harvey, S. Leybourne, P. Newbold, Testing the equality of prediction mean squared errors, Int. J. Forecast. 13 (2) (1997) 281–291.